# An Optimal Deterministic Adversary for Sorting

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**Abstract**

This paper describes an optimal deterministic, polynomial-time adversary for ordinary sorting. For any correct sorting algorithm, the adversary forces that algorithm to make $(1 - o(1))n \log_2 n$ comparisons. The best previously known adversary forced $\frac{37}{64} n \log_2 n$ comparisons. The adversary recursively moves items into ordered subproblems, extending the classical framework in which items move down a binary tree.

## Methodological disclosure

Except for the abstract, the text of this paper is AI-generated using GPT 6 Astra. The author is working on a clear, human-readable exposition.

## 1 Introduction

The information-theoretic lower bound for sorting has a simple adversarial interpretation. Keep the set of total orders consistent with the answers given so far, and answer each comparison so as to retain a larger half of that set. Starting from $n!$ orders, this adversary forces at least $\log_2(n!)$ comparisons. Its direct implementation requires counting linear extensions of a partial order, a #P-complete problem [7]. The question is whether one can obtain the same leading constant with an efficiently computable adversary.

We consider $n$ distinct labelled keys. A sorting algorithm chooses pairs adaptively, and must be correct for every strict total order on the labels. An adversary is *legal* if every finite sequence of its answers is consistent with at least one such order. It sees the requested comparisons, but need not know the sorting algorithm that generates them. Throughout the paper, log denotes $\log_2$, and ln the natural logarithm. The cases $n \leq 1$ are immediate; we assume $n \geq 2$.

**Theorem 1.1.** *There is a deterministic legal adversary, initialized using only $n$, that performs a fixed polynomial number of bit operations per comparison and forces every correct comparison sorter to make*

$$(1 - o(1))n \log n$$

*comparisons. The term $o(1)$ tends to zero as $n \to \infty$.*

Merge sort uses at most $n \lceil \log n \rceil$ comparisons, so the leading coefficient is optimal. Our construction is asymptotic: its parameter choice establishes a uniform polynomial bound, but is not intended for practical running times.

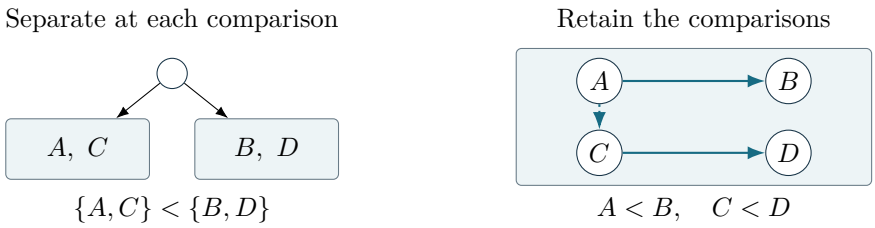



**Figure 1:** Two answered comparisons need not determine all cross-relations. The binary-tree rule separates the pairs spatially. A group with stored arrows retains the freedom to answer later cross-comparisons. The dashed arrow shows how a subsequent answer $A < C$ can be stored in the same group.

### 1.1 From binary trees to retained comparisons

Atallah and Kosaraju [1] gave an efficient sorting adversary with leading coefficient $1/2$. Related adversaries occur in the work of Borodin, Guibas, Lynch, and Yao [2] on searching after partial sorting, Richards and Vaidya [3] on the distribution of comparisons among keys, and Brodal, Chaudhuri, and Radhakrishnan [4] on maintaining the minimum. The binary-tree rule of Borodin et al. and Brodal et al. provides the starting point for our construction.

Place every item at the root of an ordered binary tree. Items in disjoint subtrees have the left-to-right order of those subtrees. A comparison changes the placement as follows.

- If both items occupy the same node, move one to its left child and the other to its right child, and answer accordingly.
- If one item is at an ancestor of the other, move the higher item to the child whose subtree does not contain the lower item.
- If their nodes are unrelated by ancestry, answer from their left-to-right order, without moving either item.

Each comparison increases the sum of item depths by at most two. At completion, occupied nodes must be distinct and unrelated by ancestry: otherwise two represented items can occur in either relative order, leaving an unresolved comparison. Their binary addresses are therefore prefix-free. If their depths are $d_1, \dots, d_n$, the corresponding dyadic intervals are disjoint, giving

$$\sum_i 2^{-d_i} \le 1, \qquad 2^{-\frac{1}{n}\sum_i d_i} \le \frac{1}{n}\sum_i 2^{-d_i} \le \frac{1}{n}.$$

Convexity gives the bound on $2^{-\frac{1}{n}\sum_i d_i}$. Hence $\sum_i d_i \ge n \log n$, and at least $\frac{1}{2} n \log n$ comparisons are necessary [4, Section 4.1].

The tree makes the state easy to maintain, but commits to some relations that have never been queried. After answers $A < B$ and $C < D$ at the same node, for example, it may place $A, C$ in the left child and $B, D$ in the right. This also fixes $A < D$ and $C < B$. Figure 1 contrasts this separation with retaining just the two inequalities.

Kaplan, Zamir, and Zwick [5] improved the leading coefficient to $37/64$ by retaining more structure before moving items down the tree. Their constructions allow ordered pairs within a node, together with rules for resolving interactions between these pairs. The present adversary extends this approach: a node retains a directed graph of comparisons, and an item moves to a child only when the graph must be simplified or its coordinate must be prepared for a query.

### 1.2 The construction and its analysis

For a partial order $P$, the region of unit coordinates satisfying its inequalities has volume $e(P)/n!$, where $e(P)$ is the number of linear extensions: every coordinate order occupies volume $1/n!$ [6].

Thus retaining a larger-volume side would implement the ideal adversary if we kept all items at the root and never simplified its constraints.

In our representation each group has auxiliary real positions in $(0, 1)$, constrained by its stored comparisons. An arrow $x \to y$ means $X_x < X_y$. The adversary chooses an approximately larger-volume answer, retaining at least $1/2 - \eta$ of the group's feasible volume. Bounded degree and bounded directed-path length allow this probability to be estimated from a neighborhood whose size depends only on an accuracy parameter. The group itself may contain arbitrarily many items.

To simplify a group, fix an item's position at an interior point $c$, replace its incident arrows by interval bounds, and give it a fresh coordinate in a child at $c$. This preserves every answered comparison. If $f_x(c)$ is the density of the fixed coordinate, the product $Z$ of all group volumes is multiplied by exactly $f_x(c)$. A well-chosen position can therefore recover volume while removing constraints from the graph.

The proof assigns a small interval to each child and embeds the entire recursive state in the unit cube. The widths of these intervals contribute a factor $\mathcal{W}$. Initially $n!Z\mathcal{W} = n!$; when a sorter finishes, the embedding lies in one order region and $n!Z\mathcal{W} \le 2^n$. We then use the potential

$$\Psi = n!Z^{\beta}\mathcal{W}\,2^{-\tau e},$$

where $e$ is the number of stored arrows, $\beta = 1 + o(1)$, and $\tau = o(1)$. The extra volume factor and the charge for stored arrows make every complete repair nondecreasing in $\Psi$. Each comparison reduces $\log \Psi$ by at most $1 + o(1)$, giving the theorem.

Section 2 specifies the algorithm before its analysis. Sections 3 and 4 establish legality and the local numerical guarantees. Sections 5 and 6 give the packing and potential arguments, and Section 7 proves the uniform polynomial bound. The proofs of the density bounds, locality, and exact local volume calculation appear in the appendices. The manuscript source is available online.[1]

## 2 The adversary

The adversary maintains a rooted tree of groups. A group records comparisons among items whose relative order is still being resolved there. An item may leave a group by fixing its position and continuing in a child at that position. We call this operation a *refinement*.

Fix an integer $k \ge 16$. The algorithm uses the parameters

$$\eta = \frac{1}{100k}, \qquad H = 2^{2k}, \qquad B = (k+1)^{2H+1}, \qquad R = 2^{2kB}. \tag{1}$$

Between comparison calls, the graph in each group has degree less than $k$ and no directed path on $H$ vertices. These two restrictions will make the probability calculations local. The threshold $1/R$ determines when a queried coordinate must first be refined. Section 7 selects $k$ as a function of $n$; for now it is fixed.

### 2.1 The state of a group

At each node $v$, every item represented there is either *free* or belongs to a child subtree. A free item $x$ has an auxiliary coordinate $X_x$ in an open interval $(a_x, b_x) \subseteq (0, 1)$. These coordinates describe possible orders; they are not the unknown key values. A directed acyclic graph on the free items stores inequalities $X_x < X_y$. Each child has a fixed position $c \in (0, 1)$, its *anchor*, and its own copy of the unit interval. Different children of one node have different anchors.

[1] https://github.com/nicknash/sorting-adversary-proofs/tree/main/for-humans

Each original item has exactly one free coordinate in the entire tree. At every ancestor of its node it is represented by the anchor of the child containing it. Initially all $n$ items are free at the root, their intervals are $(0, 1)$, and there are no arrows or children.

The feasible region of a node is

$$K_v = \{X : a_x < X_x < b_x \text{ for all free } x, \quad X_x < X_y \text{ for every arrow } x \to y\}. \tag{2}$$

Write $V_v = \mathrm{vol}(K_v)$, with volume one when the node has no free items, and define

$$Z = \prod_v V_v. \tag{3}$$

Thus $Z = 1$ initially. We shall prove that every reached state satisfies $0 < Z \le 1$. The algorithm never calculates this product. It calculates probabilities only in small induced subgraphs, as specified in Section 4.

Only actual comparisons create arrows. The graph is simple: repeated comparisons do not create parallel arrows, and transitive consequences are not inserted. The number of stored arrows will matter in the proof.

We maintain all interval bounds implied by the arrows. Write $y \preceq x$ if $y = x$ or a directed path leads from $y$ to $x$. The *effective interval* of $x$ is $(\ell_x, r_x)$, where

$$\ell_x = \max_{y \preceq x} a_y, \qquad r_x = \min_{y \succeq x} b_y, \qquad W_x = r_x - \ell_x. \tag{4}$$

The operation TIGHTEN replaces every stored interval by its effective interval. It is performed after every change to an interval or arrow, before testing widths or calculating probabilities. The resulting bounds remain stored when arrows are later removed. A coordinate is *wide* if $W_x \ge 1/R$, and *narrow* otherwise.

### 2.2 Choosing an answer

To compare original items $A$ and $B$, descend while both belong to the same child. At the first node where their representations differ, each is either a free coordinate or a child anchor. Two different anchors determine the answer. Otherwise at least one representation is free.

Suppose first that both are free. Under the uniform volume measure on $K_v$, let

$$p = \mathbb{P}(X_A < X_B) = \frac{\mathrm{vol}(K_v \cap \{X_A < X_B\})}{V_v}.$$

The probability notation is only a ratio of volumes; the adversary makes no random choices. Given an estimate $\widehat{p}$ with $|\widehat{p} - p| \le \eta$, it answers $A < B$ if $\widehat{p} \ge 1/2$, and $B < A$ otherwise. Since equality has volume zero, the chosen answer retains at least $1/2 - \eta$ of $V_v$. It is stored as an arrow. For a free coordinate and an anchor $c$, write $x$ for the free item and use $\mathbb{P}(X_x < c)$. Store the resulting upper or lower interval bound, reversing the reported label order when the free item is $B$.

The local estimator requires a wide coordinate for a threshold query, and at least one wide endpoint for a two-coordinate query. If this condition fails, the adversary refines a queried item before answering.

### 2.3 Choosing an anchor and refining an item

Every anchor is the midpoint of a dyadic cell:

$$c = (j + \tfrac{1}{2})2^{-m}, \qquad 0 \le j < 2^m, \quad m \ge 0. \tag{5}$$

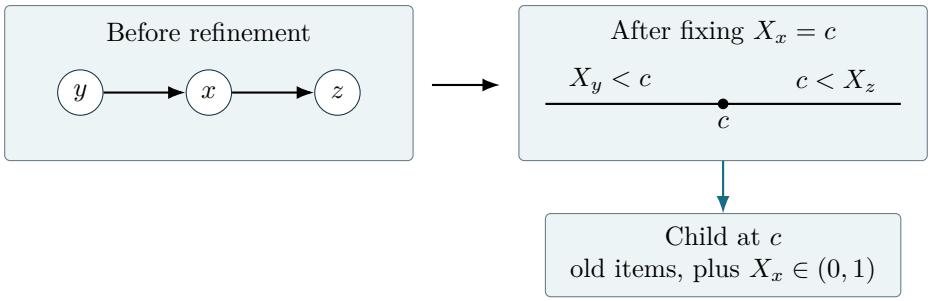



**Figure 2:** A refinement preserves the comparisons incident to $x$ as interval bounds in the parent. The relative order of $x$ and the items already at the same anchor is resolved inside the child.

We call $m$ its *address length*: $m$ binary digits specify the cell. The reduced denominator of $c$ is $2^{m+1}$, so its position determines $m$ uniquely. Reusing an anchor therefore also reuses its address length.

There are two rules for choosing $c$.

1. For a wide item, start with the cell $[0, 1]$. At its midpoint $c$, estimate $F_x(c) = \mathbb{P}(X_x < c)$ to error at most $\eta$. If the estimate is below $1/4$, retain the right half; if it is above $3/4$, retain the left half; otherwise accept $c$. Repeat within the retained cell.
2. For a narrow item, choose the dyadic length $b = 2^{-m}$ satisfying $W_x/8 < b \leq W_x/4$. Choose the midpoint of a cell of this length nearest to $(\ell_x + r_x)/2$, taking the smaller midpoint in case of a tie.

Both choices lie strictly inside the effective interval. Their density guarantees are proved in Section 4.

To refine $x$ at $c$, fix $X_x = c$, remove $x$ from the parent's free graph, and replace each incident arrow by the corresponding bound on its other endpoint. Thus $y \to x$ becomes $X_y < c$, and $x \to z$ becomes $X_z > c$. Retain all other stored bounds and tighten the parent. Then admit $x$ to the child at $c$, creating the child if necessary, as an isolated coordinate with interval $(0, 1)$. The child's previous data stay unchanged.

## 2.4 The complete comparison procedure

The following procedures specify all state changes. Ties between eligible items are resolved by label order. A path is selected by the lexicographically first eligible label sequence. Degree counts neighbors without regard to edge direction; path length below counts vertices.

**Restore**$(v)$**.**

1. While the free graph of $v$ has a vertex of degree at least $k$, refine such a vertex.
2. While it contains a directed path on $H$ vertices, select one such path and refine all its vertices in label order. Complete this entire list before selecting another path.

**Compare**$(A, B)$**.**

1. Descend through common children to the first node $v$ where the representations differ. If both are anchors, return their order.
2. Tighten $v$. If none of the free queried coordinates is wide, refine one of them and restart step 1.
3. Estimate the probability of $A < B$ to additive error $\eta$ and select an answer by the rule above. Store its arrow or interval bound.
4. Tighten $v$, run RESTORE$(v)$, and return the selected answer.

Every answer is also entered in a separate comparison graph on the original labels, before restoration or, for an anchor answer, before returning. This graph records the public transcript. Its edges do not count toward the degree restrictions or the active arrow count used below. It will supply a compatible total order for the finite stopping rule in Section 7.

# 3 Legality and the graph invariants

We first establish that the state always represents orders consistent with the public transcript. We then check that each call terminates with the graph conditions required for the next probability calculation.

## 3.1 The orders represented by a state

Choose a feasible point in each $K_v$, avoiding equalities between free coordinates and between a free coordinate and an anchor. Read the positions from left to right, replacing each child anchor by the order read recursively inside that child. This gives a strict total order on the original labels. The excluded equalities have volume zero.

Our invariant is that *every represented order respects every answered comparison.* We do not require every order consistent with the transcript to remain represented. A positive-volume state represents at least one order, so this invariant suffices for legality.

**Lemma 3.1** (Effective intervals and sections)**.** *For an acyclic arrow graph, tightening leaves the feasible region unchanged. The region has positive volume if and only if every effective width is positive. Moreover, fixing $X_x = c$ at any $c \in (\ell_x, r_x)$ leaves a positive-volume section in the remaining coordinates.*

*Proof.* Along a directed path, every feasible assignment obeys the lower bound at its first vertex and the upper bound at its last. Thus all bounds introduced by (4) already hold. The reverse inclusion follows because each effective interval is contained in its original interval.

Positive volume requires positive widths. Conversely, suppose all widths are positive. Along an arrow $x \to y$, both $\ell_x \le \ell_y$ and $r_x \le r_y$. Choose numbers $0 < \theta_x < 1$ increasing in a topological order and set

$$X_x = (1 - \theta_x)\ell_x + \theta_x r_x.$$

This expression is nondecreasing in each endpoint and strictly increasing in $\theta_x$. Every arrow is therefore strict. A sufficiently small open neighborhood of this point remains feasible, proving positive volume.

Fixing $x$ at $c$ introduces upper bound $c$ at its predecessors and lower bound $c$ at its successors. A predecessor $y$ retains positive width because $\min(r_y, c) > \ell_y$: indeed $r_y > \ell_y$ and $c > \ell_x \ge \ell_y$. The successor argument is symmetric. No other effective interval changes. Apply the positive-volume criterion to the remaining graph. □

**Lemma 3.2** (Refinement and child admission)**.** *Suppose a state satisfies the order invariant. Refining $x$ at an interior anchor $c$ preserves it, including when the child at $c$ already exists. The transcript relates $x$ to none of the labels already in that child subtree. If $f_x(c)$ is the density of the parent's coordinate $X_x$, then*

$$\frac{Z_{\text{after}}}{Z_{\text{before}}} = f_x(c). \tag{6}$$

*Proof.* Fix a feasible assignment of the remaining free parent coordinates in the section $X_x = c$, pairwise distinct and away from all child anchors. Keeping those coordinates fixed, one may move $x$ slightly to either side of $c$: all its incident inequalities are strict. The resulting represented orders put $x$ immediately before or immediately after the entire old child subtree at $c$. Both

respect the public transcript. Hence the transcript relates $x$ to none of the labels in that subtree, even by transitivity.

Inserting $x$ at any position among the child labels is consequently legal. Their mutual order is unchanged; there is no public comparison between $x$ and one of them; and $x$ stays on the same side of every outside label. This applies to every assignment in the new state, proving the invariant. The section has positive volume by Lemma 3.1, and adding an isolated unit coordinate preserves positivity.

Let $S_x(c)$ be the section volume. By definition, $f_x(c) = S_x(c)/V_v$. The parent's volume is replaced by $S_x(c)$, and the child gains an independent interval of length one. All other factors in $Z$ stay unchanged, giving (6). □

The identity uses a section of one lower dimension in the parent and a new dimension in the child. It does not condition on an event of positive probability. For example, in $0 < X_A < X_B < 1$, the density of $X_A$ at $c$ is $2(1 - c)$. Refining $A$ at $1/4$ multiplies $Z$ by $3/2$. Such increases will pay for the loss caused by simplifying a group.

### 3.2 Bounds during a comparison call

**Lemma 3.3** (Graph bounds and termination)**.** *During a comparison call, every active graph has degree at most $k$ and height at most $2H$. At its end, every graph has degree less than $k$ and height less than $H$. Restoration performs at most $n$ refinements, all at the node changed by the answer.*

*Proof.* Assume the strict bounds hold before the call. An answer adds at most one arrow, so it raises any degree by at most one. Any new directed path consists of two old paths joined by that arrow, and therefore has fewer than $2H$ vertices. A selected answer cannot create a directed cycle, since its feasible region has positive volume. Interval changes add no edges.

Every refinement removes a vertex and its incident arrows from the parent. Thus degrees and heights cannot increase during restoration. The degree pass establishes the degree bound, and the path pass preserves it while establishing the height bound. Each refinement removes a different free item from this node, so there are at most $n$. A child receives an isolated coordinate; its old graph is unchanged. No child requires restoration. □

**Lemma 3.4** (Query preparation)**.** *Steps 1–2 of* Compare *perform at most two refinements. Each selected coordinate is narrow, and preparation preserves the graph bounds.*

*Proof.* A preparation refines one of the two original queried items. Its new coordinate is isolated and has interval $(0, 1)$. Before an answer is stored, any further preparation only removes a parent coordinate and adds an isolated coordinate in a child. It cannot restrict this fresh coordinate. If the query descends into that child, the prepared item is therefore still wide. Each queried item can require preparation at most once. The selection test ensures narrowness, and deleting vertices or admitting isolated ones preserves the graph bounds. □

Consequently a call makes at most $n + 2$ refinements. Throughout the call, every item has at most $B = (k + 1)^{2H+1}$ other comparable items: each lies within undirected distance $2H$ in a graph of degree at most $k$. These are precisely the bounds used in the anchor rules and the local estimator. The argument also explains why active arrows and the public comparison graph must remain separate.

**Proposition 3.5** (Legality)**.** *Every reached state has positive volume and represents only orders consistent with the public transcript. After every complete call, all nodes satisfy the strict degree and height bounds.*

*Proof.* We induct on comparison calls. Initially the root region is the unit cube, every order is represented, and the graph is empty. Suppose the claims hold at the start of a call. Preparation preserves the graph bounds and makes a queried coordinate wide by Lemma 3.4. Each refinement uses an interior anchor and preserves positive volume and the order invariant by Lemma 3.2.

The local estimator of Section 4 now applies to the queried region. Its selected answer retains at least $1/2 - \eta > 0$ of the volume, and every remaining represented order satisfies that answer. In particular, a new arrow cannot create a directed cycle. An answer between anchors is already true in all represented orders.

Lemma 3.3 bounds the degree and height throughout restoration. Thus the numerical guarantees continue to apply to every refinement, and Lemma 3.2 preserves the order invariant. Restoration terminates with the strict graph bounds. This completes the induction. The numerical results used here are proved for arbitrary positive interval-and-arrow regions with the stated graph bounds, independently of this induction. □

# 4 Local probabilities and the choice of an anchor

The two numerical tasks are to estimate a comparison probability and to fix a coordinate where its density is sufficiently large. The graph bounds make the first task local. The second task links the volume recovered by a refinement to the address length of its child.

## 4.1 Local probability estimates

For an interval-and-arrow region, degree and distance refer to the underlying undirected graph; height is the maximum number of vertices on a directed path. Two vertices are comparable if a directed path joins them in either direction.

**Theorem 4.1** (Local interval probabilities)**.** *Fix integers $d, h \geq 1$, rational $R \geq 1$, and rational $0 < \varepsilon < 1/4$. There is a computable integer $\rho = \rho(d, h, R, \varepsilon)$ with the following property. Consider a positive-volume interval-and-arrow region in the unit cube, with degree at most $d$ and height at most $h$.*

1. *If $W_x \geq 1/R$, the probability $\mathbb{P}(X_x < c)$ is determined to additive error $\varepsilon$ by the induced radius-$\rho$ neighborhood of $x$.*
2. *If at least one of $W_x, W_y$ is at least $1/R$, the probability $\mathbb{P}(X_x < X_y)$ is determined to additive error $\varepsilon$ by the subgraph induced by the union of their radius-$\rho$ vertex neighborhoods.*

*In each case the local region retains the effective intervals of the full region. All other widths may be arbitrarily small and positive. The local region has at most*

$$b = 2(d+1)^{\rho+1}$$

*coordinates. With dyadic input bounds, its probabilities can be computed exactly in $(3b)^b \operatorname{poly}(b)$ arithmetic operations.*

The locality proof is in Appendix B. It couples conditional coordinate updates in the full and local regions. Bounded degree and height make a sufficiently high moment of the coordinate discrepancy contract. A density bound at the wide queried coordinate then converts closeness of coordinates into closeness of comparison probabilities. The proof uses random variables only to compare distributions; the algorithm calculates the local volumes deterministically by the enumeration in Appendix C.

For the adversary, use $d = k$, $h = 2H$, and $\varepsilon = \eta$. Lemmas 3.3 and 3.4 ensure the required graph and width bounds at every call to the estimator. The same local region may be reused for all thresholds in one midpoint search.

### 4.2 Density and dyadic address length

We use two facts about the density $f_x$ of a coordinate with effective interval $(\ell, r)$, width $W$, and at most $B$ other comparable coordinates. First,

$$f_x(t) \le \frac{B+1}{W}, \qquad f_x(t) \ge \frac{4^{-B}}{W} \quad \left(\ell + \frac{W}{4} \le t \le r - \frac{W}{4}\right). \tag{7}$$

Second, if $F_x$ is its cumulative distribution, then for $a \le c \le b$,

$$(b-a) f_x(c) \ge \min\{F_x(c), 1 - F_x(c)\}(F_x(b) - F_x(a)). \tag{8}$$

Appendix A proves both facts by maps between sections of the feasible region. The first controls the maximum density and gives a lower bound near the center of the effective interval. The second gives a density lower bound at a point with substantial mass on both sides.

**Lemma 4.2** (Midpoint search)**.** *The midpoint search for a wide item terminates at an interior anchor $c$. If its accepted cell has length $2^{-m}$, then*

$$f_x(c) \ge 2^{m-4}.$$

*The search makes at most $\lceil \log(4(B+1)/W_x) \rceil + 1$ tests.*

*Proof.* The estimates have error at most $\eta < 1/32$. If the current cell is $[a, b]$, every rejected test ensures $F_x(a) < 9/32$ and $F_x(b) > 23/32$. These inequalities hold initially as well. At acceptance,

$$\frac{7}{32} \le F_x(c) \le \frac{25}{32}, \qquad F_x(b) - F_x(a) \ge \frac{7}{16}.$$

Thus $c$ is interior, and (8) gives $2^{-m} f_x(c) \ge 49/512 > 1/16$. Finally, (7) implies

$$\frac{7}{16} \le F_x(b) - F_x(a) \le \frac{B+1}{W_x}\, 2^{-m}.$$

At $m \ge \lceil \log(4(B+1)/W_x) \rceil$, the upper bound is at most $1/4 < 7/16$, a contradiction. The search therefore stops within the stated number of tests. □

**Lemma 4.3** (Refinement density)**.** *Every anchor chosen by the adversary is interior and satisfies*

$$\log f_x(c) \ge (1 - 1/k)m - 4. \tag{9}$$

*For a narrow item its address length also satisfies*

$$m > 2kB + 2. \tag{10}$$

*Proof.* For a wide item, Lemma 4.2 gives the stronger lower bound $m - 4$. For a narrow item, write $b = 2^{-m}$ for the chosen cell length. Its midpoint is at distance at most $b/2 \le W_x/8$ from the interval center, so it belongs to the central half. By (7) and $W_x < 8b$,

$$f_x(c) \ge \frac{4^{-B}}{W_x} > 2^{m-3-2B}.$$

Also $b \le W_x/4 < 1/(4R)$, whence $m > \log R + 2 = 2kB + 2$. In particular $2B < m/k$, proving (9). □

The prescribed fine grid for a narrow item is important: it forces a long address even if the effective interval happens to contain a coarse dyadic midpoint. This is the property that pays for preparing a narrow query.

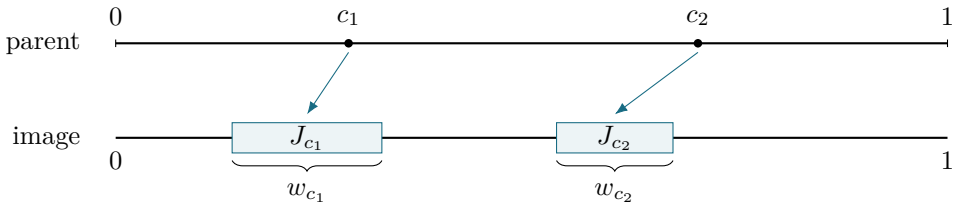



**Figure 3:** The packing map has slope 1/2 between anchors. Its jump at each anchor reserves an interval for the entire child subtree. The widths are shown schematically; their values are given by (11).

# 5 A geometric bound at termination

We now place the recursive state inside the ordinary unit cube $(0,1)^n$, with one coordinate for each original label. The map preserves every represented order. When sorting is complete its image must lie in a single coordinate-order region, of volume $1/n!$. Computing the volume of the image will therefore give an upper bound on the final state.

## 5.1 An interval for each child

Reserve for an anchor of address length $m$ a child interval of width

$$w(m) = \frac{2^{-(1+1/k)m}}{4k}. \tag{11}$$

At most $2^m$ anchors have that address length. Hence at every node

$$\sum_c w_c \le \frac{1}{4k} \sum_{m \ge 0} 2^{-m/k} = \frac{1}{4k(1-2^{-1/k})} \le \frac{1}{2}. \tag{12}$$

For the last inequality, convexity gives $2^{-t} \le 1 - t/2$ for $0 \le t \le 1$, and we take $t = 1/k$.

At a node, map a free position $t$ distinct from all anchors to

$$G(t) = \frac{t}{2} + \sum_{c<t} w_c. \tag{13}$$

The jump at an anchor $c$ leaves the open interval

$$J_c = \left( \frac{c}{2} + \sum_{a<c} w_a, \frac{c}{2} + \sum_{a<c} w_a + w_c \right).$$

Place the recursively mapped child state inside $J_c$, using an affine map from $(0,1)$ to $J_c$. The intervals $J_c$ are disjoint, occur in anchor order, and fit in $(0,1)$ by (12). Free positions stay in the gaps between them. All these widths are used only in the proof.

For a parent–child edge $c$, let $s_c$ be the number of original labels in the child's entire subtree, and put

$$\mathcal{W} = \prod_{\text{parent–child edges } c} w_c^{s_c}. \tag{14}$$

A comparison does not change $\mathcal{W}$. A refinement into a child of width $w$ multiplies it by $w$: the refined label crosses that edge, while its membership in every other subtree remains unchanged. This holds for both a new child and an existing child.

**Lemma 5.1** (Packing volume)**.** *The recursive map is injective and preserves the represented order. Its image $\mathcal{I} \subseteq (0,1)^n$ has volume*

$$\mathrm{vol}(\mathcal{I}) = 2^{-n} Z\mathcal{W}. \tag{15}$$

*Consequently, if the transcript determines a unique total order, then*

$$n! Z\mathcal{W} \le 2^n. \tag{16}$$

*Proof.* The map $G$ puts each free position on the correct side of every child interval. Recursion therefore preserves the order of the original labels. Anchors and widths are fixed within the state. The image coordinate of a label depends only on its one free coordinate, through a strictly increasing map, so the whole map is injective.

Partition the domain at the finitely many anchor hyperplanes. On each piece, the map of a label is affine. Its slope is $1/2$ at its free coordinate, multiplied by the widths of the child intervals containing that label along the path to the root. There are exactly $n$ factors of $1/2$, one per free coordinate. An edge width $w_c$ occurs $s_c$ times. Thus the Jacobian on every piece is $2^{-n}\mathcal{W}$. The domain, a product of the node regions, has volume $Z$. Injectivity lets us add the image volumes of the pieces, proving (15).

At completion, legality places the image in the region of the cube realizing the unique transcript order. The $n!$ coordinate-order regions have equal volume $1/n!$, since coordinate permutations preserve volume and equality hyperplanes have volume zero. Hence $2^{-n} Z\mathcal{W} \le 1/n!$. □

### 5.2 The addresses of a path

The same bound on child widths controls a path repair.

**Lemma 5.2** (Address length of a path batch)**.** *The $H$ labels of a selected directed path enter distinct children of their parent. If their address lengths are $m_1, \dots, m_H$, then*

$$\sum_{i=1}^{H} m_i \ge kH. \tag{17}$$

*Proof.* Every arrow on the selected path is an answered comparison. Its labels are therefore pairwise comparable in the public transcript, and remain so after arrows are deleted. By Lemma 3.2, two of these labels cannot be admitted to the same child. The procedure completes all $H$ refinements in this parent, so their distinct child widths have sum at most $1/2$. The arithmetic–geometric mean inequality gives

$$\prod_{i=1}^{H} w(m_i) \le (1/(2H))^H.$$

Substitute (11) and take logarithms:

$$(1 + 1/k) \sum_i m_i \ge H\big(\log H - \log(2k)\big).$$

Since $H = 2^{2k}$ and $\log(2k) \le k - 1$ for integers $k \ge 4$, the right-hand side is at least $(k+1)H$. Division by $1 + 1/k$ proves the claim. The auxiliary inequality follows from $2k \le 2^{k-1}$, true at $k = 4$ and preserved when $k$ increases by one. □

## 6 The comparison lower bound

The packing quantity $n! Z\mathcal{W}$ starts at $n!$ and ends at most $2^n$. It remains to relate its decrease to the number of comparisons. We adjust the volume exponent and charge a small amount for each stored arrow, so that every complete repair has nonnegative change.

Let $e$ be the total number of active arrows over all nodes. Set

$$L = \log(4k), \qquad \beta = \frac{k+1+L}{k-5}, \qquad \tau = \frac{4\beta + L}{k}, \tag{18}$$

and define

$$\Psi = n! Z^{\beta} \mathcal{W} 2^{-\tau e}. \tag{19}$$

Here $\beta > 1$ and $\tau > 0$. The coefficients satisfy

$$\beta(1 - 1/k) - (1 + 1/k) = \tau, \qquad 4\beta + L = k\tau. \tag{20}$$

The extra power of $Z$ rewards volume recovered by refinement. Removing an arrow also restores the factor $2^{\tau}$ charged when it was stored. The algorithm does not evaluate this potential.

**Lemma 6.1** (Refinement inequality)**.** *If a refinement selects an anchor of address length $m$ for a vertex of current degree $r$, then*

$$\log \Psi_{\text{new}} - \log \Psi_{\text{old}} \geq \tau(m + r - k). \tag{21}$$

*Proof.* Refinement multiplies $Z$ by $f_x(c)$ and $\mathcal{W}$ by $w(m)$, and deletes exactly $r$ active arrows. The child receives an isolated coordinate. Using Lemma 4.3,

$$\begin{aligned} \log \frac{\Psi_{\text{new}}}{\Psi_{\text{old}}} &= \beta \log f_x(c) - (1 + 1/k)m - L + \tau r \\ &\geq \left[\beta(1 - 1/k) - (1 + 1/k)\right] m - 4\beta - L + \tau r \\ &= \tau(m + r - k). \end{aligned}$$

The last line is (20). □

The choice of $\beta, \tau$ makes the address length and the deleted-arrow count have the same coefficient, while making the fixed cost equal to $k$ times that coefficient. The three reasons for refinement now fit the same inequality.

**Proposition 6.2** (Repair monotonicity)**.** *Each query-preparation refinement and each degree refinement preserves or increases $\Psi$. Each complete path batch preserves or increases $\Psi$.*

*Proof.* A preparation refines a narrow item, so $m > 2kB + 2 > k$ by (10). A degree refinement has $r \geq k$. In either case Lemma 6.1 gives a nonnegative change.

For a selected path, let $m_i$ be the address and $r_i$ the current degree at its $i$th refinement. Summing the refinement inequality over the whole batch gives

$$\log \frac{\Psi_{\text{after}}}{\Psi_{\text{before}}} \geq \tau \left( \sum_{i=1}^{H} m_i + \sum_{i=1}^{H} r_i - kH \right) \geq 0,$$

because Lemma 5.2 gives $\sum_i m_i \geq kH$ and every $r_i$ is nonnegative. □

An individual refinement within a path batch may decrease the potential. The proposition applies to the completed batch, which is why RESTORE finishes a selected path before making another selection or returning an answer.

**Theorem 6.3** (Fixed-parameter comparison bound)**.** *For every integer $k \geq 16$, the adversary is deterministic and legal. If a correct sorter terminates after $q$ calls, then, with $a_\eta = -\log(1/2 - \eta)$,*

$$q \geq \frac{\log(n!) - n}{\beta a_\eta + \tau} = \frac{n \log n - O(n)}{1 + O(\log k / k)}. \tag{22}$$

*Proof.* The procedures and tie rules are deterministic. Legality is Proposition 3.5, and termination of each call follows from Lemmas 3.3 and 3.4.

An answer between free coordinates or between a free coordinate and an anchor retains at least $1/2-\eta$ of its node volume. It leaves $\mathcal{W}$ unchanged and adds at most one active arrow. Its change in $\log\Psi$ is therefore at least $-\beta a_\eta-\tau$. An answer between anchors leaves the potential unchanged. Preparations and complete restorations do not decrease it by Proposition 6.2. Summing over calls yields

$$\log\Psi_{\text{final}} \geq \log(n!) - q(\beta a_\eta + \tau).$$

Initially $Z=\mathcal{W}=1$ and $e=0$, explaining the term $\log(n!)$. At termination, Lemma 5.1, $Z\leq 1$, and $e\geq 0$ give

$$\Psi_{\text{final}} = (n!Z\mathcal{W})Z^{\beta-1}2^{-\tau e} \leq 2^n. \tag{23}$$

Combining the last two inequalities proves the first bound.

Finally, (18) gives $\beta = 1+O(\log k/k)$ and $\tau = O(\log k/k)$. Since $\eta = 1/(100k)$, $a_\eta = 1+O(1/k)$. The estimate $\log(n!) = n\log n - O(n)$ follows, for example, by comparing $\sum_{j=1}^{n}\log j$ with $\int_1^n \log t\,dt$. These estimates give the second expression in (22). □

# 7 A uniform polynomial-time implementation

For each fixed $k$, the local calculation has bounded dimension, but this bound grows with $k$. We choose $k=k(n)$ slowly enough that its entire calculation is polynomial in $n$, while $k(n)$ still tends to infinity. We measure computation on a fixed deterministic multitape Turing machine, with integers represented in binary.

## 7.1 Selecting the parameter

For a parameter $k$, let

$$\rho_k = \rho(k, 2H, R, \eta), \qquad b_k = 2(k+1)^{\rho_k+1}, \qquad E_k = (3b_k)^{b_k}.$$

The formula for $\rho$ in Appendix B is computable by integer arithmetic. Fix a program that successively writes the complete parameter tuples for $k=16,17,\ldots$, including $H, B, R, \rho_k, b_k, E_k$ in binary. At initialization simulate this program for $n$ steps. Among the completed tuples, choose the largest $k$ satisfying

$$E_k \leq n. \tag{24}$$

If there is none, answer every comparison according to label order.

The simulation produces $O(n)$ bits, and testing its completed outputs takes polynomial time. For each fixed $k$, its tuple is eventually written and $E_k$ is finite. Thus, for all sufficiently large $n$, that tuple has been produced within the budget and satisfies (24). The selected $k(n)$ consequently tends to infinity.

The gate bounds the number of orders enumerated in any local volume calculation by $n$. It remains to bound the lengths of the integers used in those calculations and the number of calculations per call.

## 7.2 Precision and state size

Every interval endpoint is 0, 1, or a previously chosen dyadic anchor. Let

$$K_0 = \lceil \log(4(B+1)R) \rceil + 2.$$

Lemma 4.2 bounds the number of fractional binary digits in an anchor chosen for a wide item by $K_0$. Since the initialization budget includes writing $B$ and $R$, we have $K_0 = O(n)$.

Suppose the existing endpoints have at most $\Lambda$ fractional digits. A positive difference between two such endpoints is at least $2^{-\Lambda}$. For a narrow item, its chosen cell length satisfies $2^{-m} > W_x/8$, so $m < \Lambda + 3$ and its midpoint has at most $\Lambda + 3$ fractional digits. Tightening only copies endpoints and introduces no new precision. After $P$ refinements, therefore,

$$\Lambda \le K_0 + 3P. \tag{25}$$

By the termination bounds, $P \le (n+2)q$ after $q$ complete calls, and the next call performs at most $n+2$ more refinements. There are at most $P+1$ nodes.

To bound state size even for indefinitely repeated queries, after

$$Q_{\text{stop}} = 2n\,\lceil \log(n+1) \rceil \tag{26}$$

complete calls, choose a total order extending the saved public comparison graph and answer from it thereafter. A topological ordering gives such an order by legality. The recursive state can then be discarded. The public graph contains every earlier answer, including answers between anchors, so this change remains consistent with the entire transcript.

Before the cutoff, (25) gives $\Lambda = O(n^2 \log(n+1))$ and the tree has polynomially many nodes. All thresholds in a wide-item midpoint search have at most $K_0$ fractional digits as well. By Appendix C, a local calculation on $b$ coordinates uses integers of

$$O\big(b(\Lambda + \log(b+1))\big)$$

bits. Its enumeration has at most $n$ terms by (24), and each term requires a polynomial number of integer operations. Standard binary arithmetic therefore gives a fixed polynomial bit bound.

A wide-item refinement makes at most $K_0$ such calculations; a narrow-item refinement uses only its endpoints. There are at most $n+2$ refinements per call and one final probability estimate. Tightening, neighborhood searches, and path selection are polynomial graph computations. In particular, longest path lengths in a directed acyclic graph are found in reverse topological order; these lengths allow the lexicographically first path on $H$ vertices to be selected greedily. Thus the bit cost per comparison is bounded by one polynomial in $n$, whose exponent is independent of $k$.

*Proof of Theorem 1.1.* The parameter selection and finite cutoff define one deterministic polynomial-time adversary. For all sufficiently large $n$, it chooses a parameter, and $k(n) \to \infty$. If sorting ends before the cutoff, Theorem 6.3 gives $q \ge (1 - o(1))n \log n$. If it reaches the cutoff, it has already made more than $n \log n$ comparisons. Legality holds throughout, including after freezing the compatible order. This proves the theorem. □

# A Coordinate densities

We prove the density bounds used to choose anchors and to turn coordinate approximation into probability approximation. Throughout this section, density means normalized section volume.

**Lemma A.1** (Coordinate density bounds)**.** *Suppose $x$ has at most $B$ other comparable labels and effective interval $(\ell, r)$ of width $W$. Its density satisfies*

$$f_x(t) \le \frac{B+1}{W}, \tag{27}$$

$$f_x(t) \ge \frac{4^{-B}}{W} \qquad (\ell + W/4 \le t \le r - W/4). \tag{28}$$

*These bounds also hold after fixing any set of labels incomparable with $x$, for every resulting slice of positive volume.*

*Proof.* Let $S(t)$ be the section volume at $X_x = t$, with any specified incomparable labels already fixed. For $\ell < t < s < r$, put $\alpha = (r-s)/(r-t)$ and apply

$$\phi(z) = \begin{cases} s + \alpha(z-t), & t \le z \le r, \\ z, & z \ge r \end{cases}$$

to $x$ and all its descendants. Every descendant has value greater than $t$ and upper endpoint at least $r$. The map moves these coordinates upward, keeps their interval constraints, and preserves strict inequalities within this set because it is strictly increasing. Any arrow crossing the set's boundary enters it, so that arrow also remains strict.

This injective map sends the section at $t$ into the section at $s$. In the section's free coordinates its derivatives are $\alpha$ or 1, with at most $B$ affected coordinates. Partitioning at the finitely many thresholds $X_y = r$ therefore gives

$$S(s) \ge \left(\frac{r-s}{r-t}\right)^B S(t) \qquad (t < s). \tag{29}$$

For $\ell < s < t < r$, apply to $x$ and its ancestors the increasing map

$$\phi(z) = \begin{cases} z, & z \le \ell, \\ s + \dfrac{s-\ell}{t-\ell}(z-t), & \ell \le z \le t. \end{cases}$$

Their lower endpoints are at most $\ell$; they move downward, and every boundary arrow leaves this set. The same argument yields

$$S(s) \ge \left(\frac{s-\ell}{t-\ell}\right)^B S(t) \qquad (s < t). \tag{30}$$

Both maps fix incomparable coordinates. From any nonempty section they reach every point of $(\ell, r)$, so fixing those coordinates does not change the support of $x$.

Normalize $S$ to obtain $f_x$. Integrating the two inequalities on their respective sides of $t$ gives

$$1 \ge f_x(t)\left(\int_\ell^t \left(\frac{s-\ell}{t-\ell}\right)^B ds + \int_t^r \left(\frac{r-s}{r-t}\right)^B ds\right) = \frac{W f_x(t)}{B+1}.$$

This proves the upper bound. Some section has density at least $1/W$. Moving it to any $t \in [\ell + W/4, r - W/4]$ uses a ratio at least $1/4$ in (29) or (30), proving the lower bound. □

**Corollary A.2** (A bound for small differences)**.** *If $y$ is incomparable with $x$, or $Y$ is a fixed number, then for every fixed interval $I$ of length $a$,*

$$\mathbb{P}(X_x - Y \in I) \le \frac{(B+1)a}{W}, \tag{31}$$

*where $Y = X_y$ in the first case.*

*Proof.* In the first case, condition on $X_y$. Lemma A.1 bounds the conditional density of $x$ by $(B+1)/W$. Integrating over the translate $I + X_y$ and then averaging over $X_y$ proves the claim. For fixed $Y$, apply the unconditional density bound. □

**Lemma A.3** (Balancing sections)**.** *Let $f$ be a coordinate density, extended by zero outside its effective interval. For $u \le t \le v$,*

$$f(u)f(v) \le f(t)f(u+v-t). \tag{32}$$

*Proof.* Write $S$ for the unnormalized section volume. The claim is immediate if $t = u$, $t = v$, or either outer section is empty. Otherwise take feasible assignments $p, q$ in the sections $X_x = u, v$, and put $\delta = t - u$. Subtract $\delta$ from every coordinate of $q$ to obtain $h$. Both $p$ and $h$ satisfy the strict arrows.

Form an undirected graph by joining the endpoints of an arrow $i \to j$ when

$$\max(p_i, h_i) \geq \min(p_j, h_j).$$

Along each such edge, $h - p$ is nonzero and has the same sign at both endpoints. Indeed, if the signs were different or one difference were zero, the strict inequality in $p$ or in $h$ would contradict the displayed condition. At $x$ the difference is $v - t > 0$, so $h_i > p_i$ throughout the connected component containing $x$.

Swap $p, h$ on that component, then add $\delta$ back to the second assignment. The output coordinates at $x$ are $u + v - t$ and $t$. At each swapped coordinate,

$$p_i < h_i = q_i - \delta < q_i, \qquad p_i < p_i + \delta < q_i,$$

so both new values remain in the original interval. Internal arrows remain strict. For a boundary arrow, the absence of a joining edge means that either value at its lower endpoint is less than either value at its upper endpoint. Thus the outputs are feasible.

The map is injective. Subtracting $\delta$ from the second output recovers the same unordered coordinate pairs, hence the same graph and component; swapping that component again recovers $p, h$. On each of finitely many pieces, the map permutes free coordinates and adds constants, so it preserves volume. Consequently

$$S(u)S(v) \leq S(t)S(u + v - t).$$

Dividing by the square of the total volume proves the claim. □

**Lemma A.4** (Density from mass on both sides)**.** *For a coordinate density $f$, its cumulative distribution $F$, and $a \leq c \leq b$,*

$$(b - a)f(c) \geq \min\{F(c), 1 - F(c)\}\big(F(b) - F(a)\big). \tag{33}$$

*Proof.* Lemma A.3 gives $f(u)f(v) \leq f(c)f(u + v - c)$ for $u \leq c \leq v$. Integrate over $u \leq c$ and $c \leq v \leq b$. For each fixed $v$, the integral of the shifted density is at most one, giving

$$F(c)\big(F(b) - F(c)\big) \leq (b - c)f(c).$$

Integration over $a \leq u \leq c$ and $v \geq c$ similarly gives

$$\big(F(c) - F(a)\big)\big(1 - F(c)\big) \leq (c - a)f(c).$$

Add these inequalities and replace the two tail factors on the left by their minimum. □

# B Local approximation of comparison probabilities

We prove Theorem 4.1 by coupling the uniform distributions on the full feasible region and the retained region. A sequence of coordinate updates preserves each distribution and reduces the discrepancy between coordinates away from the retained boundary.

### B.1 Coupled coordinate updates

Start with a uniform assignment in each region, under any joint coupling. Visit the full graph in a topological order; the retained graph uses the restriction of this order. Conditional on every other coordinate, $X_x$ is uniform on $(a, b)$, where

$$a = \max\{\ell_x,\ X_y : y \to x\}, \qquad b = \min\{r_x,\ X_z : x \to z\}.$$

Replace it by $(1 - U_x)a + U_x b$. Use independent uniform $U_x \in (0, 1)$ at each vertex and in each sweep, sharing the same $U_x$ between the two assignments at retained vertices. Vertices outside the retained graph are updated only in the full assignment. Each update resamples the conditional distribution, and hence preserves the corresponding joint uniform law.

Let $\Delta_x$ be the absolute difference between the two values at a retained vertex. Whenever every incident edge is retained,

$$\Delta_x^{\text{new}} \le (1 - U_x) \max_{y \to x} \Delta_y^{\text{new}} + U_x \max_{x \to z} \Delta_z^{\text{old}}. \tag{34}$$

An empty maximum is zero. The bound follows because taking a maximum or minimum with common fixed bounds does not increase the largest discrepancy among its arguments. Predecessors have already been updated; successors have not.

Suppose the undirected ball $B_h(x)$ is retained. Recursively expand the predecessor term in (34), choosing a maximizing predecessor at each step. Any predecessor chain reaches a vertex with no predecessor within at most $h$ vertices. Define $A_x$ as the minimum of the fresh $1 - U_y$ over $x$ and its ancestors. The coefficient following that chain to a zero discrepancy is at least $A_x^h$. At each expansion the two coefficients sum to one; every other branch ends at an old successor discrepancy, reached by at most $h - 1$ predecessor edges and one successor edge. Thus

$$\Delta_x^{\text{new}} \le (1 - A_x^h) \max_{y \in B_h(x)} \Delta_y^{\text{old}}. \tag{35}$$

This bound holds for every choice of maximizing predecessors, since $A_x$ uses all ancestors. All edges used by the expansion lie in $B_h(x)$.

### B.2 Moment contraction

Put

$$M = (d + 1)^{h+1}, \qquad a = \frac{1}{4M^2}, \qquad p = (4M^2)^{h+1}.$$

The ball $B_h(x)$ and the set of ancestors defining $A_x$ each have at most $M$ vertices. Hence

$$\mathbb{P}(A_x < a) \le Ma = \frac{1}{4M}.$$

On the complementary event, Bernoulli's inequality gives

$$(1 - A_x^h)^p \le (1 - a^h)^p \le \frac{1}{1 + pa^h} = \frac{1}{1 + 4M^2} \le \frac{1}{4M}.$$

Here $(1 - z)^{-p} \ge 1 + pz$ for $0 \le z < 1$. It follows that

$$\mathbb{E}[(1 - A_x^h)^p] \le \frac{1}{2M}.$$

The fresh random variables defining $A_x$ are independent of all old discrepancies. Raising (35) to the power $p$ and bounding the maximum by a sum therefore yields

$$\begin{aligned} \mathbb{E}[(\Delta_x^{\text{new}})^p] &\le \frac{1}{2M} \sum_{y \in B_h(x)} \mathbb{E}[(\Delta_y^{\text{old}})^p] \\ &\le \frac{1}{2} \max_{y \in B_h(x)} \mathbb{E}[(\Delta_y^{\text{old}})^p]. \end{aligned} \tag{36}$$

### B.3 The retained radius

Choose

$$C = M + 1, \qquad \delta = \frac{\varepsilon}{16CR},$$
$$\lambda = \lceil \log(1/\delta) \rceil, \qquad \mu = \lceil \log(8/\varepsilon) \rceil,$$
$$t = p\lambda + \mu, \qquad \rho = ht.$$

These integers are computable by rational arithmetic and comparisons with powers of two.

Fix a queried vertex $x_*$. After sweep $j$, apply (36) only at vertices within distance $h(t-j)$ of $x_*$. Their required $h$-balls lie within distance $h(t-j+1)$, where the preceding sweep's bound applies. Every initial discrepancy is at most one because all coordinates lie in $(0,1)$. Induction on $j$ gives

$$\max_{x \in B_{h(t-j)}(x_*)} \mathbb{E}[\Delta_x(j)^p] \le 2^{-j}, \qquad 0 \le j \le t.$$

In particular, Markov's inequality gives

$$\mathbb{P}(\Delta_{x_*}(t) > \delta) \le 2^{-t}\delta^{-p} \le 2^{-\mu} \le \varepsilon/8. \tag{37}$$

For two queried vertices, apply this estimate at each one.

### B.4 Comparison probabilities

If the queried labels are comparable, their directed path has at most $h-1$ edges and is retained because $\rho \ge h$. The full and local comparison probabilities are then the same, either zero or one.

Otherwise the labels are incomparable. Let $x$ be a queried coordinate of width at least $1/R$. It has at most $M$ other comparable labels, so Corollary A.2 bounds the probability that $X_x - X_y$ lies in any interval of length $a$ by $CRa$. Outside the two exceptional events in (37), a difference in comparison signs between the two assignments requires

$$|X_x - X_y| \le 2\delta.$$

The probability of this event in the full assignment is at most $4CR\delta = \varepsilon/4$. The exceptional events together have probability at most $\varepsilon/4$, so the signs disagree with probability at most $\varepsilon/2$. For a threshold comparison, the same argument uses one exceptional event and an interval of length $2\delta$. Since each assignment retains its original uniform law, the difference between the two probabilities is at most their probability of disagreement.

Finally, the retained region has positive volume: the restriction of a full strictly feasible assignment has a small feasible neighborhood. The union of radius-$\rho$ neighborhoods of two vertices has at most $2(d+1)^{\rho+1}$ vertices. This proves the approximation and size claims of Theorem 4.1.

## C Exact local volume

Let the retained graph have $b$ coordinates. We compute its volume by partitioning its feasible assignments according to the order of the coordinates and their interval endpoints.

### C.1 Enumeration of orders

Sort the distinct interval endpoints as

$$e_1 < e_2 < \cdots < e_s, \qquad s \le 2b.$$

Enumerate the total orders of these endpoints and the $b$ labelled coordinates that preserve the endpoint order. Insert the coordinates successively into the ordered list. The $j$th insertion has $s + j$ positions, so the number of orders is at most

$$\prod_{j=1}^{b}(s+j) \le (3b)^b.$$

Reject an order if it places a coordinate outside its interval or reverses an arrow. For an accepted order, let $r_j$ count the coordinates between $e_j$ and $e_{j+1}$. Their relative order is fixed. Permuting these coordinates partitions their cube into $r_j!$ equal-volume regions, apart from equalities of volume zero. The accepted order therefore has volume

$$\prod_{j=1}^{s-1}\frac{(e_{j+1}-e_j)^{r_j}}{r_j!}. \tag{38}$$

An empty gap contributes one. The interval check excludes coordinates outside the extreme endpoints, so $\sum_j r_j = b$.

Almost every feasible assignment determines exactly one accepted order, and every accepted order describes only feasible assignments. Summing (38) thus gives the volume. For $b = 0$, the volume is one.

### C.2 Integer arithmetic and probability queries

Let $Q = 2^\Lambda$ be a common denominator of the dyadic endpoints, and write $d_j = Q(e_{j+1} - e_j)$. Every summand in (38) has denominator $b!Q^b$ and integer numerator

$$\frac{b!}{\prod_j r_j!}\prod_j d_j^{r_j}. \tag{39}$$

The factorial quotient is an integer because it counts partitions of $b$ labelled positions into groups of sizes $r_j$. Sum the numerators and divide by $b!Q^b$ once at the end.

All coordinates lie in $(0, 1)$, so the total volume is at most one. The summands are nonnegative; every partial numerator sum is therefore at most $b!Q^b$. The factors used to form one numerator also have products at most this bound, since $d_j \le Q$ and $\sum_j r_j = b$. Thus the arithmetic uses integers of

$$O\big(b(\Lambda + \log(b+1))\big)$$

bits. Checking an order and calculating its contribution require a polynomial number of arithmetic operations in $b$. The total is $(3b)^b \operatorname{poly}(b)$ operations, proving the remaining claim of Theorem 4.1.

For a comparison numerator, add its arrow and repeat the calculation. For a threshold $c \le \ell_x$, the numerator is zero; for $c \ge r_x$, it equals the denominator. Otherwise replace the queried coordinate's upper endpoint by $c$ and repeat. There are still at most $2b$ endpoints. Use a common $Q$ for numerator and denominator; the factors $b!Q^b$ cancel. Inconsistent constraints yield no accepted order and hence volume zero.

If the resulting probability is $N/D$ with nonnegative integers $N, D$ and $D > 0$, the answer test is $2N \ge D$. The midpoint tests use $4N < D$ and $4N > 3D$. The stage gate (24) bounds the number of enumerated orders by $n$, and the preceding bit bound completes the arithmetic estimate used in Section 7.

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